

Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Switch Statement and Menu-Driven Programming in C
Lab Number	07
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab 07: Switch Statement and Menu-Driven Programming in C

Objective

To understand and implement the switch-case statement in C programming for decision-making and to develop menu-driven programs using user-defined functions.

Software Requirements

- Code::Blocks IDE / Dev-C++
- GCC Compiler.
- Windows Operating System.

Experiments Performed

In-Lab Tasks

1. Determine whether a number is even or odd using the switch-case statement.
2. Develop a simple calculator using switch-case to perform arithmetic operations.
3. Calculate grace marks based on the student's class and the number of failed subjects.

Post-Lab Tasks

1. Develop a menu-driven program to perform:
 - Factorial of a number
 - Prime number checking
 - Even/Odd checking
 - Exit from the program

Learning Outcomes

After completing this lab, the student will be able to:

- Use the switch-case statement for multiple decision-making.
- Implement menu-driven programs.
- Create and call user-defined functions.
- Perform mathematical and logical operations.
- Improve program modularity and user interaction.

In-lab Tasks:

Task1:

Determine whether a number is even or odd using the switch-case statement.

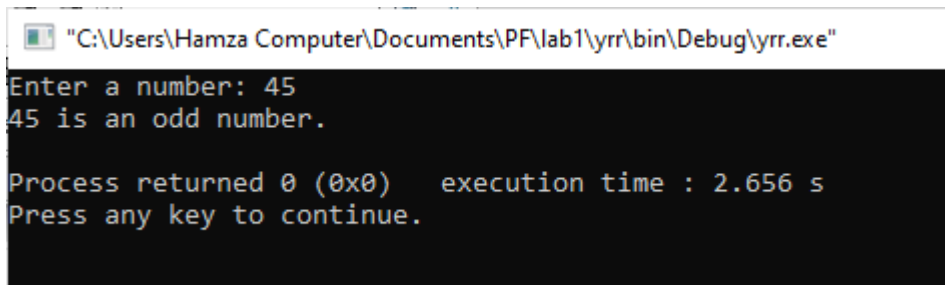
Code:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int number;
    printf("Enter a number: ");
    scanf("%d", &number);

    switch (number%2)
    {
        case 0:
            printf("%d is an even number.\n", number);
            break;
        case 1:
            printf("%d is an odd number.\n", number);
            break;
    }
    return 0;
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\yrr\bin\Debug\yrr.exe"
Enter a number: 45
45 is an odd number.

Process returned 0 (0x0)   execution time : 2.656 s
Press any key to continue.
```

Task2:

Develop a simple calculator using switch-case to perform arithmetic operations.

Code:

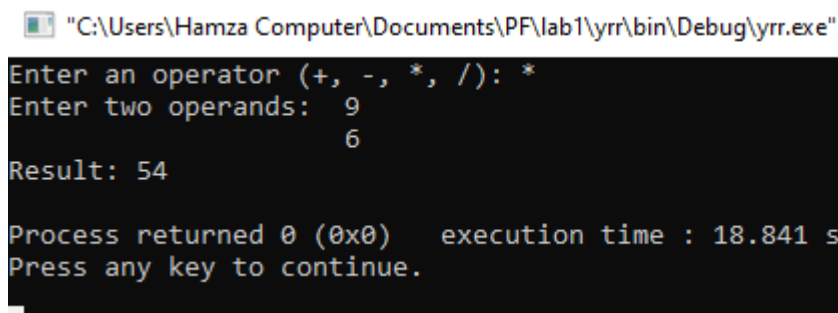
```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    char operator;
    int x, y, result;
    printf("Enter an operator (+, -, *, /):");
    scanf(" %c", &operator);
    printf("Enter two operands: ");
    scanf("%d %d", &x, &y);
    switch (operator)
    {
        case '+':
            result= x + y;
            printf("Result: %d\n", result);
            break;
        case '-':
            result= x - y;
            printf("Result: %d\n", result);
            break;
        case '*':
            result= x * y;
            printf("Result: %d\n", result);
            break;
        case '/':
            if (y != 0) {
                result= x / y;
                printf("Result: %d\n", result);
            } else {
```

```

        printf("Error: Division by zero is not allowed.\n");
    }
    break;
default:
    printf("Error: Invalid operator\n");
    break;
}
return 0;
}

```

Output:



```

"C:\Users\Hamza Computer\Documents\PF\lab1\yrr\bin\Debug\yrr.exe"
Enter an operator (+, -, *, /): *
Enter two operands: 9
                   6
Result: 54

Process returned 0 (0x0)   execution time : 18.841 s
Press any key to continue.

```

Task 03:

Calculate grace marks based on the student's class and the number of failed subjects.

Code:

```

#include <stdio.h>
#include <stdlib.h>
int main()
{
    int Class, subjects_Failed;
    int grace_Marks = 0;
    printf("Enter the class obtained 1, 2 & 3: ");
    scanf("%d", &Class);
    printf("Enter the number of subjects failed: ");

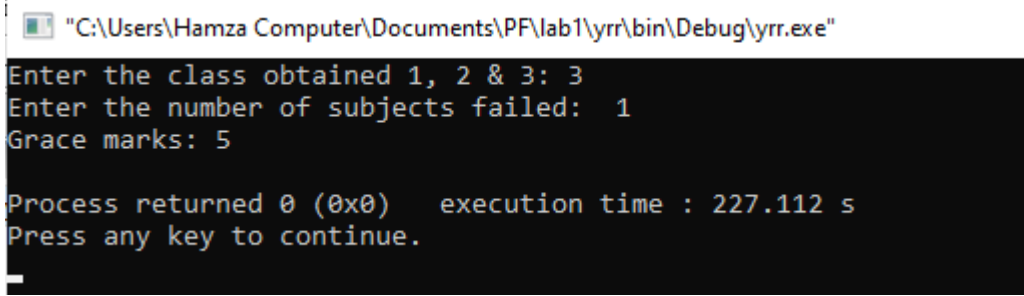
```

```

scanf("%d", &subjects_Failed);
switch (Class) {
    case 1:
        if (subjects_Failed <= 3) {
            grace_Marks = subjects_Failed * 5;
        } break;
    case 2:
        if (subjects_Failed <= 2) {
            grace_Marks = subjects_Failed * 5;
        } break;
    case 3:
        if (subjects_Failed == 1) {
            grace_Marks = subjects_Failed * 5;
        } break;
    default:
        printf("Invalid class obtained.\n");
        return 0; }
printf("Grace marks: %d\n", grace_Marks);
return 0;
}

```

Output:



```

"C:\Users\Hamza Computer\Documents\PF\lab1\yrr\bin\Debug\yrr.exe"
Enter the class obtained 1, 2 & 3: 3
Enter the number of subjects failed: 1
Grace marks: 5

Process returned 0 (0x0)   execution time : 227.112 s
Press any key to continue.

```

Post-Lab Tasks:

Task 01:

Develop a menu-driven program to perform:

- Factorial of a number.
- Prime number checking.
- Even/Odd checking.
- Exit from the program.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    int choice, num;
    do {
        printf("Menu:\n");
        printf("1. Factorial of a number\n");
        printf("2. Prime or not\n");
        printf("3. Odd or even\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter a number:");
                scanf("%d", &num);
                printf("Factorial of %d is %d\n", num, factorial(num));
                break;
            case 2:
                printf("Enter a number:");
```

```

        scanf("%d", &num);
        if (Prime(num))
            printf("%d is a prime number\n", num);
        else
            printf("%d is not a prime number\n", num);
        break;
    case 3:
        printf("Enter a number:");
        scanf("%d", &num);
        if (Even(num))
            printf("%d is an even number\n", num);
        else
            printf("%d is an odd number\n", num);
        break;
    case 4:
        printf("Exiting the program...\n");
        break;
    default:
        printf("Invalid choice! Please select a valid option.\n");
    }
    printf("\n");
} while (choice!= 4);
return 0;
}

int factorial(int num)
{
    int fact= 1, i;
    for (i=1; i<=num; i++) {
        fact*= i; }
    return fact;
}

```

```

int Prime(int num)
{
    int i;
    if (num <= 1)
        return 0;
    for (i=2; i<=num/2; i++) {
        if (num%i ==0)
            return 0; }
    return 1;
}

```

```

int Even(int num)

```

```

{
    if (num%2 ==0)
        return 1;
    else
        return 0;
}

```

Output:

 "C:\Users\Hamza Computer\Documents\PF\lab1\fy\bin\Debug\fy.exe"

```

Menu:
1. Factorial of a number
2. Prime or not
3. Odd or even
4. Exit
Enter your choice: 1
Enter a number: 120
Factorial of 120 is 0

Menu:
1. Factorial of a number
2. Prime or not
3. Odd or even
4. Exit
Enter your choice: 3
Enter a number: 86
86 is an even number
}

```